

INTERNATIONAL GCSE PHYSICS

9203/2

Paper 2

Mark scheme

November 2024

Version: 1.0 Final



2 4 B Y 9 2 0 3 / 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	C		1	AO1 3.4.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	particles move in different directions		1	AO1 3.4.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	radiation		1	AO3 3.3.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	conduction		1	AO3 3.4.2a
	convection		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	metals contain free electrons	allow delocalised electrons	1	AO1 3.4.2a
	that move through the metal or that transfer energy quickly		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	any one from: - metals are strong - the cup will not easily become damaged - metals are malleable		1	AO3 3.4.2

Total Question 1		8
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	bat		1	AO2 3.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	800 (Hz)		1	AO3 3.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	longitudinal		1	AO1 3.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	$330 = 60\,000 \times \lambda$		1	AO2 3.3.1h
	$\lambda = \frac{330}{60\,000}$		1	
	$\lambda = 0.0055 \text{ (m)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	frequency (of ultrasound wave) is above 20 000 Hz		1	AO3 3.3.3a
	so frequency is too high for humans to hear		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	any one from: • (prenatal) scanning / imaging • removal of kidney stones		1	AO1 3.3.3h

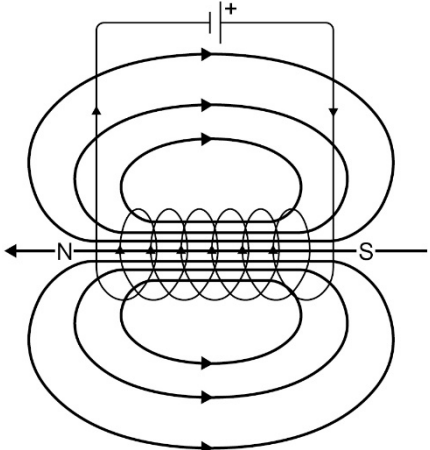
Total Question 2		9
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	force A = magnetic		1	AO1 3.3.1a,d
	force B = weight		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	(the arrows) are equal in length	allow size for length	1	AO1 3.1.3g
	(the arrows) are opposite in direction		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	$F = 0.20 \times 9.8$	allow 2.0 (N)	1	AO2 3.1.1e
	$F = 1.96 \text{ (N)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	loops above and below the solenoid	ignore any lines drawn inside the coil	1	AO1 3.5.2g
	direction of field lines		1	
				

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	(the direction of the magnetic field) reverses	allow the poles reverse	1	AO1 3.5.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	(the strength of the magnetic field) increases		1	AO1 3.5.2g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.7	decreases the magnetic force	decreases the strength of the magnetic field	1	AO3 3.1.3c,h 3.5.2e
	magnetic force on the globe is less than its weight		1	
	so globe accelerates towards the base (until magnetic force equals weight)	MP3 dependent on MP1 or MP2	1	
	OR decreases the magnetic force (1) so the height at which the resultant force is zero is lower (1) so the globe moves closer to the base (1)	MP3 dependent on MP1 or MP2		

Total Question 3		13
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	Level 3: The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.		5–6	AO4 3.5.1n
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.		3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.		1–2	
	No relevant content		0	
	Indicative content • completed circuit diagram • connect the battery, ammeter, resistor, variable resistor and switch in series • connect the voltmeter in parallel with the resistor • measure the current in the resistor using the ammeter • measure the potential difference across the resistor using the voltmeter • vary the resistance of the variable resistor • record a range of values of current and potential difference • switch circuit off between readings • reverse connections on the battery • repeat measurements of current and potential difference in negative direction • plot a graph of current against potential difference			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	to reduce heating or so that the resistance does not change	to prevent heating in the resistor	1	AO4 3.5.1i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	axes labels: y-axis = current in amps x-axis = potential difference in volts	allow voltage for potential difference	1	AO2 × 3 AO3 × 1 3.5.1i
	all points plotted correctly	allow 1 mark for 3 or 4 points plotted correctly	2	
	line of best fit		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	the line of best fit (is a straight line but) does not go through the origin		1	AO4 3.5.1i
	all of the current values are too high	allow there is a zero error	1	
	by 0.05 A		1	

Total Question 4		14
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	condition of the vehicle's brakes		1	AO1 3.1.5e
	mass of the vehicle		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	(braking distance) increases		1	AO2 3.1.5e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	braking distance of car is 20 m and braking distance of lorry is 40 m		1	AO2 3.1.5e
	the braking distance of the lorry is double that of the car		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	stopping distance includes thinking distance (and braking distance)		1	AO1 3.1.5c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	braking distance = 45 m		1	AO2 3.2.1a
	$2\,700\,000 = F \times 45$	subsequent marks may be awarded if an incorrect value read from the bar chart is used	1	
	$F = \frac{2\,700\,000}{45}$	allow a correct rearrangement using an incorrectly / not converted value of W	1	
	$F = 60\,000 \text{ (N)}$	allow a correct calculation using an incorrectly / not converted value of W	1	

Total Question 5		10
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	any two from: • surface area of the solar panel • angle of tilt of the solar panel (relative to the sun) • dirty solar panel • direction the solar panel is pointing • time of day • amount of cloud cover	allow size of solar panel allow intensity of sunlight	2	AO3 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	efficiency = $\frac{7.5}{30}$		1	AO2 3.2.2f
	efficiency = 0.25		1	
	or			
	efficiency = $\frac{7.5 \times 100}{30}$ (1)			
	efficiency = 25% (1)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	advantage any one from: - no cloud cover - can operate for more hours in a day - free up space for agriculture / housing - intensity of sunlight is greater		1	AO3 3.2.3d
	disadvantage any one from: - expensive to build - difficult to fix if there is a fault - expensive to put satellite in space		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	$3100 = \frac{\text{distance}}{86\,400}$		1	AO2 3.1.2c
	distance = $3100 \times 86\,400$		1	
	$267\,840\,000 = 2 \times 3.14 \times r$		1	
	or $r = \frac{267\,840\,000}{2 \times 3.14}$		1	
	$r = 42\,649\,681.53 \text{ (m)}$ $r = 43\,000\,000 \text{ (m)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	the gravitational / centripetal force decreases (as the distance from the Earth increases)	dependent on MP1	1	AO1 × 1 AO3 × 1 3.8.2f
	this results in a lower orbital speed		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	any one from: • share scientific ideas • reduce the cost of funding for each country • share information / resources • peer review		1	AO3 3.8.2

Total Question 6		14
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	$2.5 = \frac{1}{f}$		1	AO2 3.2.2b 3.3.1g
	$f = \frac{1}{2.5}$		1	
	$f = 0.4 \text{ (Hz)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	$h = 0.625 \text{ (m)}$	allow subsequent marks if height is incorrectly / not converted	1	AO2 3.2.1d,e
	$E = 40 \times 9.8 \times 0.625$		1	
	$E = 245 \text{ J}$		1	
	$245 = 0.5 \times 40 \times v^2$		1	
	$v = \sqrt{\frac{245}{0.5 \times 40}}$		1	
	$v = 3.5 \text{ (m/s)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	<u>work</u> done against air resistance	allow friction for air resistance	1	AO1 × 1 AO2 × 1 3.2.1b
	so energy is dissipated to the surroundings		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	$E_k = \frac{1}{2}mv^2$ and masses are different		1	AO3 3.2.1e
	but same maximum speed		1	

Total Question 7		13
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	${}_{55}^{137}\text{Cs} \longrightarrow {}_{56}^{137}\text{Ba} + {}_{-1}^0\beta$	1 mark for top row	1	AO2 3.7.2f
		1 mark for bottom row	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	number of half-lives = 4	allow $24 \times (2 \times 2 \times 2 \times 2)$	1	AO2 3.7.2h
	activity = 24×2^4		1	
	activity = 384 (Bq)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	irradiation is the exposure of an object to ionising radiation		1	AO1 3.7.1i
	contamination is the presence of (unwanted) radioactive material on an object		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	hazard	allow can cause cancer	1	AO3 3.7.2j
	causes mutation / death of cells			
	precaution	allow any sensible suggestion	1	
	any one from: • handle from a distance • use lead shielding • use a radiation detector badge			

Total Question 8		9
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